

KAIRAKUM, Tajikistan

WMO: 385980

Lat: 40.267N

Lon: 69.817E

Elev: 1138

StdP: 14.10

Time Zone: 5.00 (E05)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.8	23.3	13.3	11.3	19.7	18.4	14.4	24.8	34.9	36.7	32.1	35.1	12.2	270	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.7	97.2	71.6	94.7	70.3	92.5	69.8	75.9	91.3	74.4	89.8	72.9	88.0	8.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	119.6	87.5	69.2	111.8	85.5	67.5	105.3	83.7	40.4	91.5	38.9	90.1	37.4	88.0	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.3	25.4	22.2	15.2	103.3	7.5	2.5	9.8	105.1	5.4	106.5	1.1	107.9	-4.4	109.7		
(5)			13.9	80.9	7.3	3.0	8.7	83.0	4.4	84.7	0.3	86.4	-5.0	88.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	59.1	35.8	37.3	49.9	61.2	72.2	79.7	82.4	78.8	70.8	58.1	44.7	37.1	(6)
(7)		DBStd	17.94	5.01	7.43	7.49	6.68	4.58	4.22	3.62	4.02	4.80	7.00	6.77	5.96	(7)
(8)		HDD50	1501	441	356	98	6	0	0	0	0	0	17	182	401	(8)
(9)		HDD65	4010	906	775	470	145	3	0	0	0	9	229	608	865	(9)
(10)		CDD50	4835	0	2	93	343	689	892	1004	894	626	267	24	1	(10)
(11)		CDD65	1867	0	0	1	31	227	442	539	429	184	14	0	0	(11)
(12)		CDH74	19951	0	0	2	204	1629	4786	6860	4762	1610	98	0	0	(12)
(13)		CDH80	8916	0	0	0	42	444	2095	3579	2244	502	10	0	0	(13)
(14)	Wind	WSAvg	9.0	11.1	10.7	9.7	8.1	8.7	8.4	7.3	6.5	6.9	8.9	10.1	11.5	(14)
(15)	Precipitation	PrecAvg	6.4	0.6	0.7	1.0	1.1	0.8	0.2	0.1	0.0	0.1	0.6	0.6	0.7	(15)
(16)		PrecMax	13.0	2.3	1.9	2.9	3.3	2.7	0.9	0.9	0.4	0.9	3.5	2.4	2.4	(16)
(17)		PrecMin	3.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.2	0.5	0.4	0.7	0.8	0.6	0.2	0.2	0.1	0.2	0.7	0.5	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.5	58.9	70.6	85.4	91.8	99.3	103.1	98.0	91.8	81.4	66.8	53.6	(19)
(20)			MCWB	42.4	49.0	57.2	63.4	66.1	70.1	72.4	74.5	68.1	63.5	57.0	47.7	(20)
(21)		2%	DB	45.5	52.8	66.1	79.7	88.0	95.1	98.8	95.2	88.2	76.6	61.6	49.3	(21)
(22)			MCWB	40.6	46.4	55.7	61.3	65.8	68.0	71.8	72.0	67.3	61.8	54.1	43.8	(22)
(23)		5%	DB	43.5	49.2	62.9	75.4	84.7	92.3	95.7	93.0	85.5	72.8	58.2	46.8	(23)
(24)			MCWB	39.5	43.9	54.0	60.7	65.5	68.0	71.0	71.2	65.8	59.7	51.9	42.3	(24)
(25)		10%	DB	42.0	46.7	60.2	72.3	81.9	89.7	93.5	90.6	82.7	69.8	54.8	44.7	(25)
(26)			MCWB	38.5	42.3	52.8	59.6	65.3	68.8	70.3	70.3	64.2	58.3	49.0	41.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.1	50.2	58.9	65.8	71.2	76.2	78.3	77.7	71.8	65.6	57.7	47.9	(27)
(28)			MCDWB	47.3	57.1	67.9	78.2	82.4	89.3	92.1	93.0	86.8	77.7	65.0	53.1	(28)
(29)		2%	WB	41.4	47.2	56.6	63.4	69.4	74.3	76.6	75.1	69.3	63.3	54.9	44.8	(29)
(30)			MCDWB	44.0	52.2	63.8	75.0	82.1	87.4	92.1	90.7	84.4	74.2	61.1	48.4	(30)
(31)		5%	WB	40.0	44.5	54.8	61.8	68.2	72.8	75.1	73.4	67.6	61.2	52.3	42.7	(31)
(32)			MCDWB	42.6	48.4	60.9	71.6	80.0	86.2	91.3	89.8	82.4	70.8	57.6	45.8	(32)
(33)		10%	WB	38.8	42.4	53.4	60.7	67.0	71.4	73.5	71.7	65.9	59.4	49.6	41.3	(33)
(34)			MCDWB	41.5	46.1	59.6	70.4	78.6	84.8	89.3	87.3	79.9	67.8	54.3	44.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	10.1	13.3	17.2	18.7	19.9	22.7	23.5	24.0	18.3	13.4	8.8	(35)
(36)			MCDBR	9.2	13.1	17.3	23.1	23.6	24.7	26.8	27.1	27.2	23.4	17.3	10.4	(36)
(37)			MCWBR	6.3	8.3	8.1	9.3	8.9	9.4	10.5	11.1	11.6	11.3	10.9	6.8	(37)
(38)		5% WB	MCDBR	8.5	12.4	15.2	19.2	19.5	19.8	24.0	26.1	27.1	21.2	16.2	10.1	(38)
(39)			MCWBR	6.3	8.4	7.8	8.6	8.9	11.0	12.0	12.3	13.0	11.4	10.5	7.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.364	0.414	0.479	0.498	0.492	0.491	0.476	0.452	0.439	0.478	0.437	0.369	(40)	
(41)		taud	2.157	2.037	1.924	1.931	2.001	2.010	2.053	2.114	2.082	1.921	1.987	2.130	(41)	
(42)		Ebn at Noon	242	245	242	248	253	253	255	258	252	219	209	227	(42)	
(43)		Edh at Noon	34	44	55	58	56	55	52	48	46	49	40	32	(43)	
(44)	All-Sky Solar Radiation	RadAvg	622	876	1243	1645	2041	2355	2384	2167	1770	1155	712	538	(44)	
(45)		RadStd	61	108	109	174	128	99	81	98	58	110	90	68	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		N/A	N/A	N/A	+0.90	N/A	N/A	N/A	N/A	+155	+113

Nomenclature: See separate page